

Eric Bond | CV

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Research Interests

Broadly, my research interests span both pure (topos theory and categorical logic) and applied (verification of software and hardware, formal mathematics, implementation of theorem provers) aspects of type theory. I am a proponent of mathematically driven approaches to programming language and type theory research. My goal is to work on projects that enable the creation of the next generation of interactive theorem provers. This includes researching new core calculi, semantics for these core calculi, and implementation methods.

Education

University of Michigan
PhD Computer Science

Ann Arbor, MI
2023–Current

Purdue University
B.S. Computer Science
Minor in Mathematics

West Lafayette, IN
2013–2017

Research Experience

Two Six Technologies
Lead Research Scientist

Seattle, WA
April 2021 - August 2023

- Worked as a member of the Applied Mathematics team on several DARPA research programs including DPRIVE, SafeDocs, VSPELLS, PEARLS, and HARDEN.
- Verified optimizations for high bit width arithmetic primitives for FHE accelerators in Isabelle/HOL.
- Formalized colored operads in Coq to reason about composition of DSLs.
- Identified research opportunities and assisted principal researchers with proposal writing.
- Languages/Technologies:: Isabelle/HOL, Coq, Agda, Haskell.

Galois
Research Intern

Portland, OR
May 2019 – August 2019

- Contributed to projects aimed at delivering provably secure software via formal methods.
- Designed and implemented an automata-based API usage checker for the Crucible symbolic simulator.
- Languages/Technologies:: Haskell, LLVM, Crucible

Lawrence Livermore National Lab
Research Intern

Livermore, CA
May 2018 – August 2018

- Member of the 2018 Data Science Summer Institute.
- Participated in a quantum computing working group.
- Built a pipeline to generate simulation data for quantum chemical systems using VASP.
- Prototyped an extension of TopoMS, a tool for topological data analysis of molecular systems, which allowed for the analysis of potential fields.
- Languages/Technologies:: C++, Python, Keras, TopoMS, VASP, MPI

Quantum Information and Computation Theory Group

Undergraduate Researcher

Purdue University

August 2017 – October 2018

- Investigated mathematical formalizations for quantum programming languages for the quantum gate/circuit model.
- Surveyed various existing languages and frameworks such as Quantomatic, Quipper, QASM, QWire, Qiskit and others.

Collaborative Robotics Laboratory

Undergraduate Researcher NSF REU

Purdue University

January 2017 – June 2017

- Researched and implemented methods for task based self-adaptation reasoning using an RDF ontology (Apache Jena) and semantic networks.

Publications

○ Accepted

- E. Bond and M. Heimerdinger. Curbing the vulnerable parser: Graded modal guardrails for secure input handling. In *2023 IEEE Secure Development Conference (SecDev)*, pages 126–132, Atlanta, GA, USA, Oct 2023. IEEE Computer Society
- William Harrison, Ian Blumenfeld, Eric Bond, Chris Hathhorn, Paul Li, May Torrence, and Jared Ziegler. Formalized high level synthesis with applications to cryptographic hardware. In Kristin Yvonne Rozier and Swarat Chaudhuri, editors, *NASA Formal Methods*, pages 332–352, Cham, 2023. Springer Nature Switzerland
- Yanzhe Cui, Richard M. Voyles, Xuexuan Zhao, Jiali Bao, and Eric S. Bond. A software architecture supporting self-adaptation of wireless control networks. In *2017 13th IEEE Conference on Automation Science and Engineering (CASE)*, pages 346–351, 2017

○ Preprints

- Zachary Flores, Angelo Taranto, Eric Bond, and Yakir Forman. A formalization of operads in coq, 2023
- Zachary Flores, Angelo Taranto, and Eric Bond. A formal algebraic framework for dsl composition, 2023

○ Drafting

- Dialectica Petri Processes - Continuing work from the 2022 AMS MRC.

Presentations

- Model Validation for DARPA DPRIVE - HCSS 2022
- Polynomial Functors and Wiring Diagrams - for JHU APL under DARPA VSPILLS Jan. 2022
- Stronger Types : A Brief Introduction to Refinement Types and Dependent Types - Haskell Exchange 2020
- Mendler F-Algebras in Cedille - PurPL March 2019
- Topological Exploration of Sans-Li Potential - poster at LLNL DSSI symposium Aug. 2018
- Intro to Quantum Computation - PurPL & Purdue ACM March 2018
- Quantum Languages - presented to Sabre Kais QIQC group Feb. 2018

Projects

AMS Mathematical Research Communities: Dialectica Categories

- Member of Valeria de Paiva's group investigating applications of Dialectica Categories.
- Leading a subgroup interested in formalizations of Dialectica Categories in Agda and computational implementations in Catlab.
- Formalized Dialectica Categories and Dialectica Petri Nets in Agda
 - <https://github.com/vcvpaiva/Dialectica/tree/main>

Masters Thesis (Incomplete):

- Interactive proof assistants represent inductive data and induction principals differently depending on the underlying type theory. Coq utilizes a complicated implementation of an intrinsic calculus of inductive constructions. Cedille is a proof assistant based on a pure extrinsic type theory with a notably small core

implementation in which induction principles are generically derivable for data and computation represented as lambda encoded F-algebras.

- We investigated transformation of proof developments between these two proof assistants to study trustworthiness of formal verifications in either type theory.
- Collaborators/Advisers: Prof. Aaron Stump (Iowa), Prof. Benjamin Delaware (Purdue)

Polynomial Functors and Dynamical Systems in Cubical Agda:

- Agda implementation of Nelson Niu and David Spivak's work on polynomial functors.
- <https://github.com/bond15/Polynomials-Categorically>

MetaTheory a la Agda & Free Monads:

- Ported Meta Theory a la Carte to Agda
- Formal definition of free monads in Agda.
- <https://github.com/bond15/Metattheory-a-la-Agda>

Awards

NSF CISE CSGrad4US

Graduate Fellowship Award

August 2022

- Granted \$37k stipend and a \$16k cost-of-education allowance per year for 3 years

AMS Mathematics Research Community

Research Retreat & Travel Reimbursement

2022

- Granted \$2,500 in travel reimbursements for multiple meetings including a week-long research retreat in Beaver Hollow NY and the 2023 Joint Mathematics Meetings.

Work Experience

47 Degrees

Seattle WA

Software Engineer 2 - Functional Programming Consultant

January 2020 – April 2021

- Redesigned existing services and drove development of new functional services while teaching engineers new functional programming patterns.
- Helped multiple clients adopt Scala (a la Haskell via the Typelevel stack) for cleaner and safer development via strong typing and functional abstractions.
- Languages/Technologies:: Scala, Cats (Haskell Typeclasses in Scala), Typelevel stack, OpenAPI, AWS, GCP, Docker.

Amazon - (Alexa Machine Learning Platform)

Cambridge, MA

Software Development Engineer Intern

May 2017 – August 2017

- Designed and implemented a data access and visualization tool for scientists and engineers to view critical data for machine learning applications/training.
- Languages/Technologies:: Java, Javascript, AWS (S3, DynamoDB, Lambda), Mockito

Amazon - (Fulfillment by Amazon Technologies)

Seattle, WA

Software Development Engineer Intern

June 2016 – August 2016

- Extended an FBA seller central web applications with a widget for aggregating and displaying shipping details.
- Languages/Technologies:: Java, Javascript, Spring Framework, Mockito

Delphi Test Lab

Purdue Research Park

Software Verification Engineer Intern

May 2015 - June 2016

- Extended functionality of web based workflow and engineering standards management tool.

Teaching Experience

CS 252 - Systems Programming

Purdue University

Graduate Teaching Assistant

January 2019 – May 2019

- Taught students entry-level systems programming in C/C++ through labs on creating a web server, memory allocator, and shell.

CNIT - Introductory C Programming

Undergraduate Teaching Assistant

Purdue University

September 2014 - December 2014

- Taught students the fundamentals of programming in C.

Community

Type Theory Forall Podcast

Guest & Co-Host

2021-2022

- Episodes: 4,14,15,20

Haskell Love Conference

QA moderator & Track Manager

August 2020, September 2021